

Introduction to Computer Networking

What Is a Computer Network?

A computer network is a collection of interconnected devices, such as computers, servers, routers, and switches, that can exchange data and share resources with one another. Networks allow users to send emails, browse the web, share files, and access remote services. The internet itself is the largest example of a computer network, connecting billions of devices worldwide through a shared set of communication protocols.

Types of Networks

Networks are often classified by their geographic scope. A Local Area Network (LAN) connects devices within a small area, such as a home, office, or school building, typically using Ethernet cables or Wi-Fi. A Wide Area Network (WAN) spans large geographic areas, connecting multiple LANs across cities or countries; the internet is the most well-known WAN. A Metropolitan Area Network (MAN) covers a city or campus, falling between the size of a LAN and a WAN. Personal Area Networks (PAN), such as Bluetooth connections between a phone and headphones, cover only a few meters.

The OSI Model

The Open Systems Interconnection (OSI) model is a conceptual framework that standardizes how different networking systems communicate. It divides network communication into seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. Each layer handles a specific aspect of communication, from the physical transmission of bits over a cable in the Physical layer, to routing packets between networks in the Network layer, to delivering data to the correct application in the Application layer. Understanding the OSI model helps engineers diagnose network problems by isolating which layer is responsible for an issue.

IP Addresses and Routing

Every device on a network is assigned an IP address, a unique numerical label that identifies it and allows other devices to locate it. IPv4 addresses consist of four numbers separated by dots, such as 192.168.1.1, while the newer IPv6 standard uses a longer format to accommodate far more devices. Routers are specialized devices that forward data packets between networks based on their destination IP address, determining the most efficient path for data to travel from source to destination. This process is called routing, and it relies on routing tables and protocols such as BGP and OSPF to make forwarding decisions.

Network Protocols

Protocols are agreed-upon rules that govern how data is formatted and transmitted across a network. TCP (Transmission Control Protocol) ensures reliable, ordered delivery of data by establishing a connection and confirming receipt of packets, making it suitable for tasks like loading web pages or sending emails. UDP (User Datagram Protocol) is faster but does not guarantee delivery or order, making it useful for real-time applications like video streaming or online gaming where speed matters more than perfect accuracy. HTTP and HTTPS are application-layer protocols used for transferring web content, with HTTPS adding encryption for security via TLS.

Network Security Basics

As networks connect more devices, securing them becomes essential. A firewall monitors and controls incoming and outgoing network traffic based on predefined security rules, acting as a barrier between trusted internal networks and untrusted external ones. A Virtual Private Network (VPN) creates an encrypted tunnel between a device and a remote server, protecting data from interception when using public networks. Encryption protocols like TLS protect data in transit, while techniques such as network segmentation limit the damage an attacker can do if one part of a network is compromised.

Wireless Networking

Wi-Fi allows devices to connect to a network without physical cables, using radio waves governed by the IEEE 802.11 family of standards. Modern Wi-Fi standards, such as Wi-Fi 6 (802.11ax), offer faster speeds, better performance in crowded environments, and improved power efficiency for connected devices. Wireless networks are typically secured using WPA3 encryption, which replaced the older and less secure WPA2 standard, protecting against unauthorized access and eavesdropping.

Cloud Networking and Future Trends

As organizations move workloads to the cloud, networking has evolved to support virtualized infrastructure. Software-Defined Networking (SDN) separates the control plane from the data plane, allowing network administrators to manage traffic flow programmatically rather than through manual configuration of individual devices. Content Delivery Networks (CDNs) cache content at servers located closer to end users, reducing latency for services like video streaming. Looking forward, technologies such as 5G, edge computing, and network automation continue to shape how data moves across increasingly complex, distributed systems.